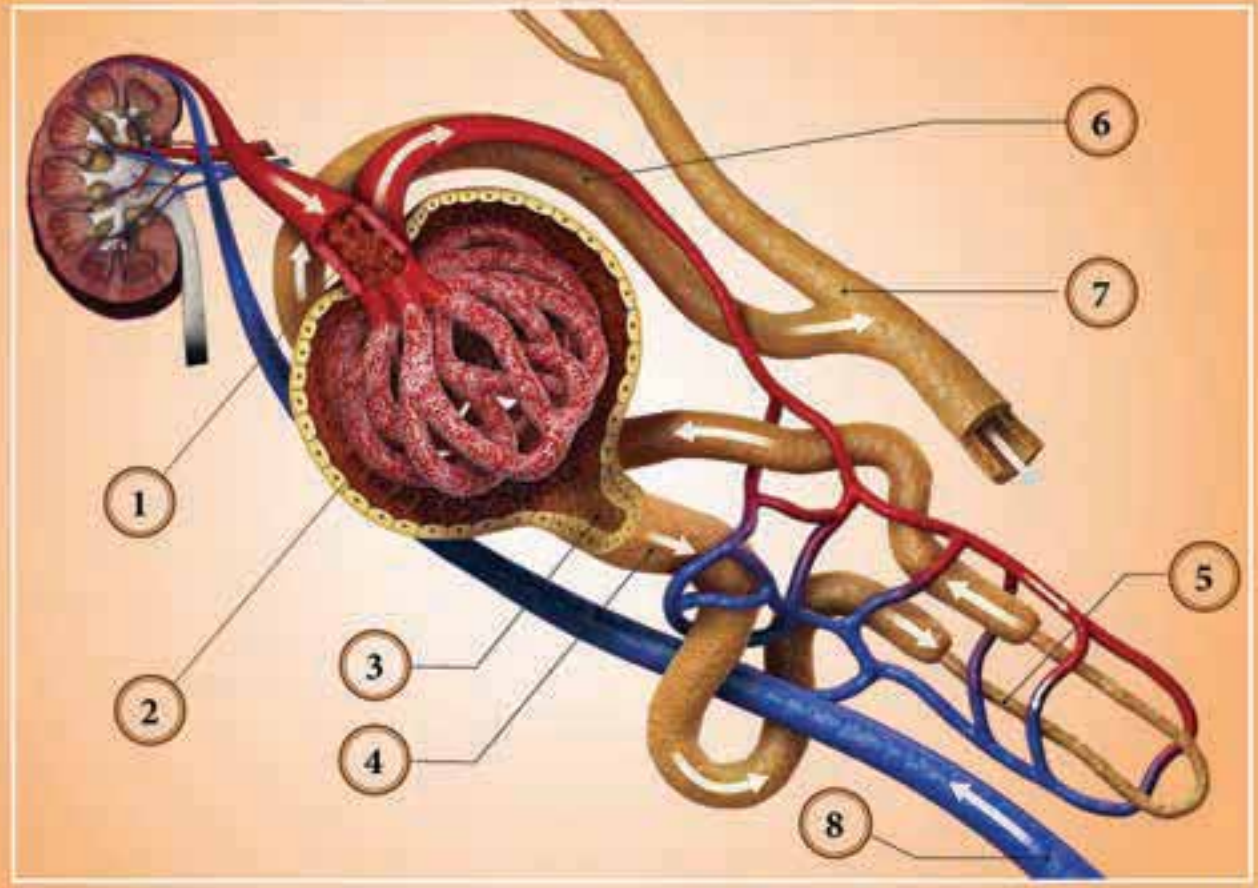
Filtration in the Kidneys



INSIDE A KIDNEY, EACH MICROSCOPIC NEPHRON IS A COMPLEX NETWORK OF COILED CAPILLARIES AND WINDING TUBES. DOZENS OF SUBSTANCES MOVE TO AND FRO TO ELIMINATE WASTES AND FINE-TUNE URINE COMPOSITION.

1 BLOOD ENTERS THE GLOMERULUS

Blood flows from a renal arteriole into the knot of capillaries. It enters at high pressure, which will force water and other substances out of the capillaries into the capsule space.

2 CAPILLARY TO CAPSULE

Blood cells and most blood proteins are too big to cross the capsular membrane into the capsule space. But the membrane's slits and pores allow through water, mineral salts, polypeptides, and other small molecules, including wastes such as urea ammonia and creatinine.

3 PROXIMAL TUBULE

Proximal (near) to its capsule, this region allows much water to be reabsorbed into the capillaries and surrounding fluids, as well as glucose, mineral salts, and other useful substances.

4 PERITUBULAR CAPILLARIES

Also called the vasa recta, this network reabsorbs up to 99 per cent of the water in the tubule, as well as various other substances. Using active pumps, it also moves sodium from the blood to the tubule.

5 HENLE'S LOOP

As the loop of Henle dips into the renal shafts, more water moves from the tubule into the blood, as well as small amounts of mineral salts and some urea and creatinine. Some acids and amines may move into the tubule, while ammonia can go in either direction.

6 DISTAL TUBULE

Distal to (far from) its capsule, this region may see water go in or out of the tubule, depending on the concentration of water already in the tubule, while hydrogen and potassium ions move to regulate both blood and urine pH. Acids, amines, and ammonia compounds may also be transported into the tubule.

7 COLLECTING DUCT

Fine adjustment of urine composition continues into the collecting-duct system. About 5 per cent of all the water and sodium being reabsorbed into the blood is recovered here.

8 VENOUS FLOW

Blood flowing away from the nephrons carries 99 per cent of its original water, 98 per cent of its sodium, calcium, and chloride, and about 40 per cent of its urea.

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